Script-Controlled ACL – Restrict Record Access Based on Field Value

Description :

This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access.

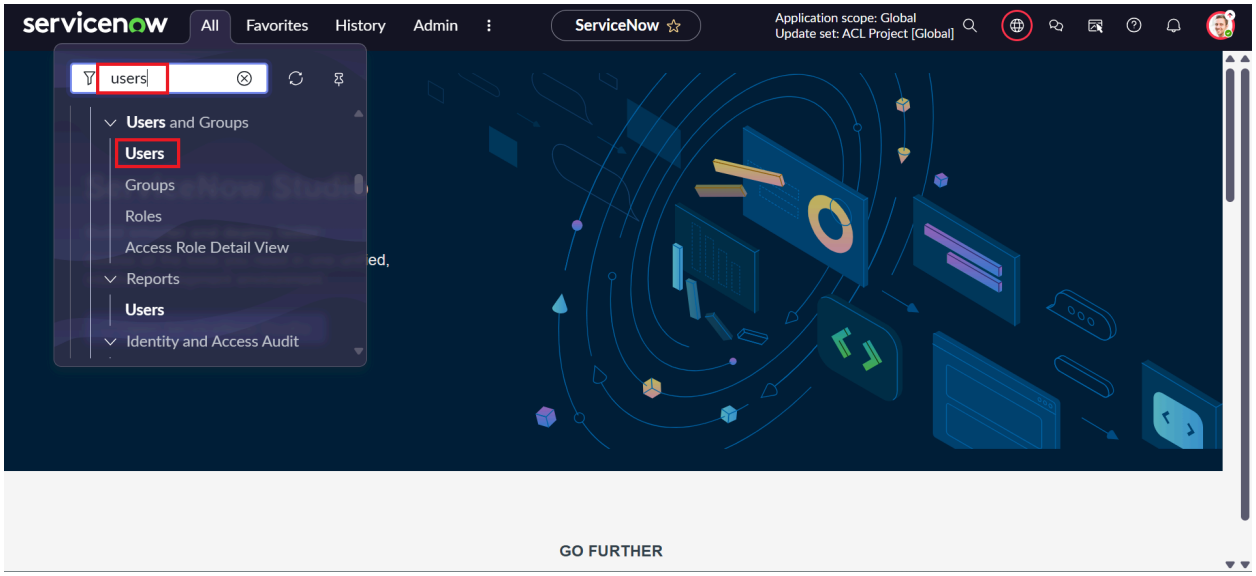
Timer :

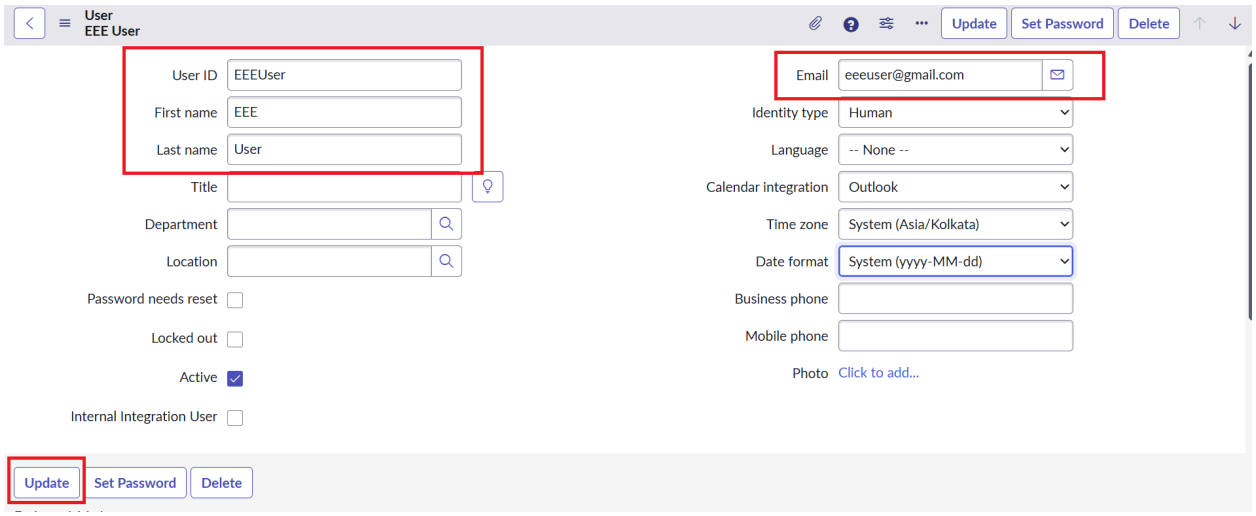
3 hrs

Milestone-1 Creation of Users and Roles

User Creation:

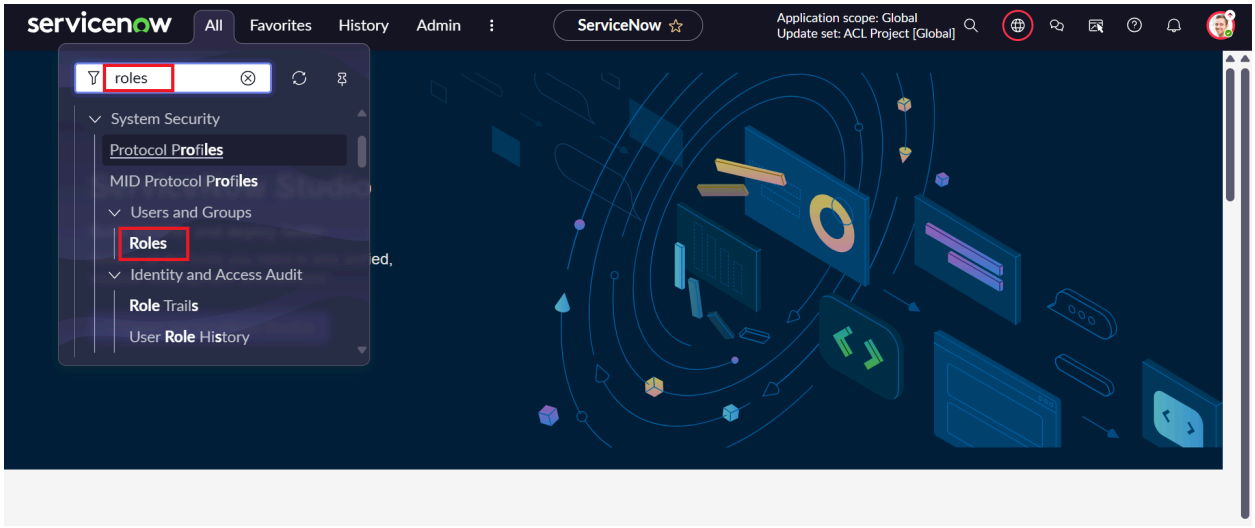
1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
    - Navigate to **User Administration → Users**
    - Click **New**
    - **User ID**: EEE User
    - **First name**: EEE
    - **Last name**: User
    - **Email**: eeeuser@gmail.com
    - Save the record

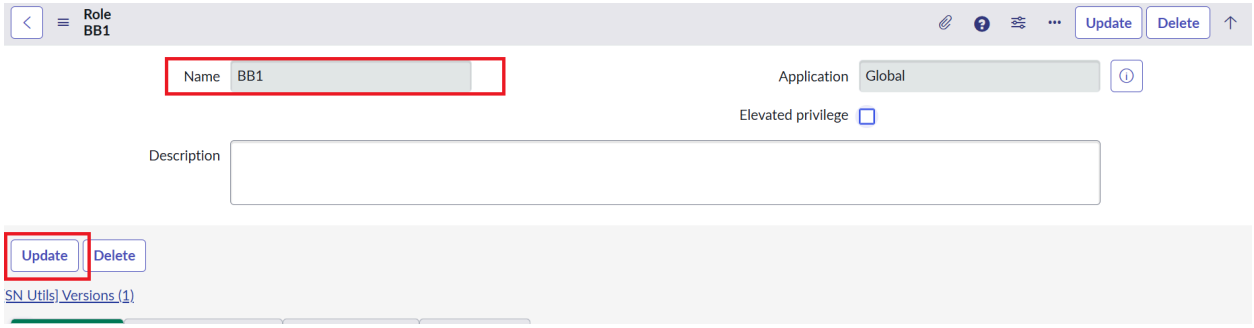




Role Creation

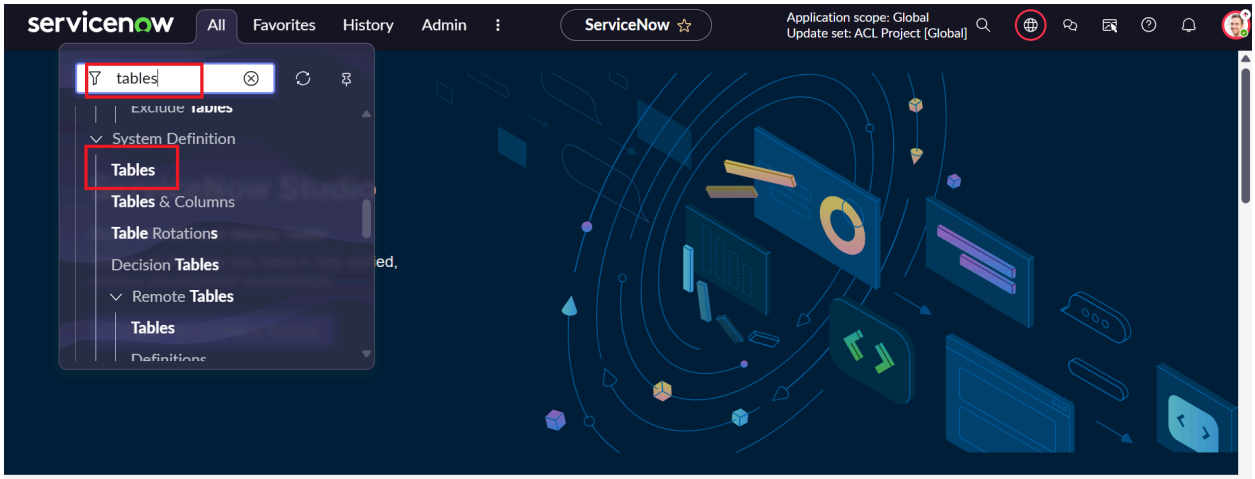
- Navigate to **User Administration → Roles**
- Click **New**
- **Name**: Enter role name as bb1
- **Description**: Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User

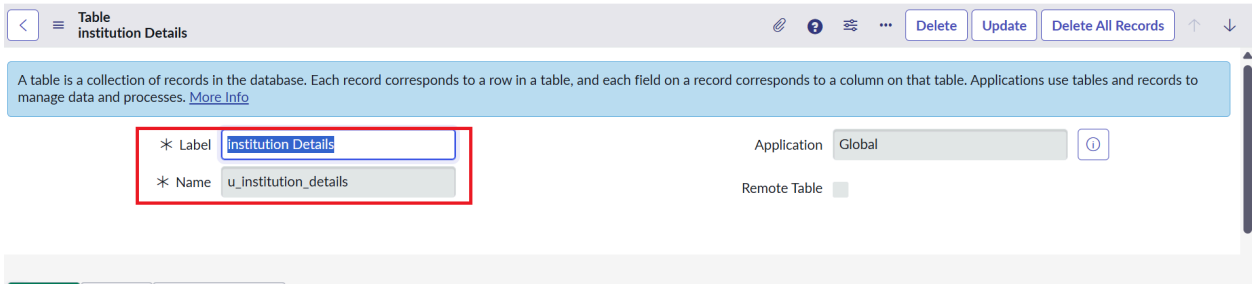




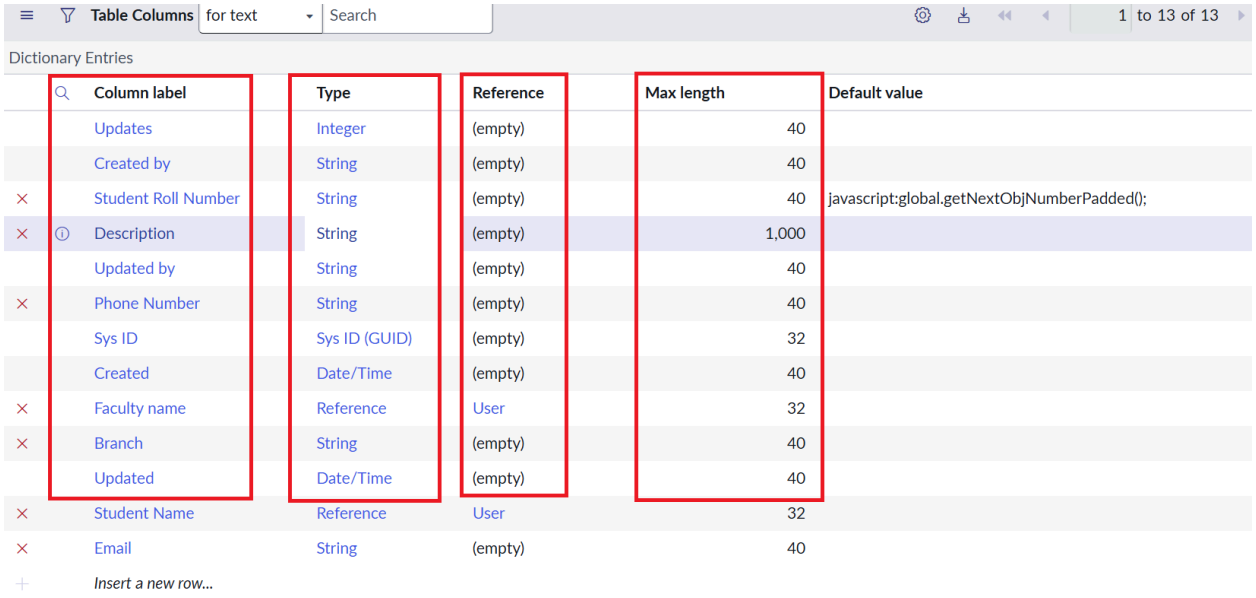
Tables Creation

1. In the **Application Navigator**, search for **System Definition → Tables**.
2. Click **New** and create a custom table with the following details:
    - **Label:** Institution Details
    - **Name:** u_institution_details
    - **Extends for :** false

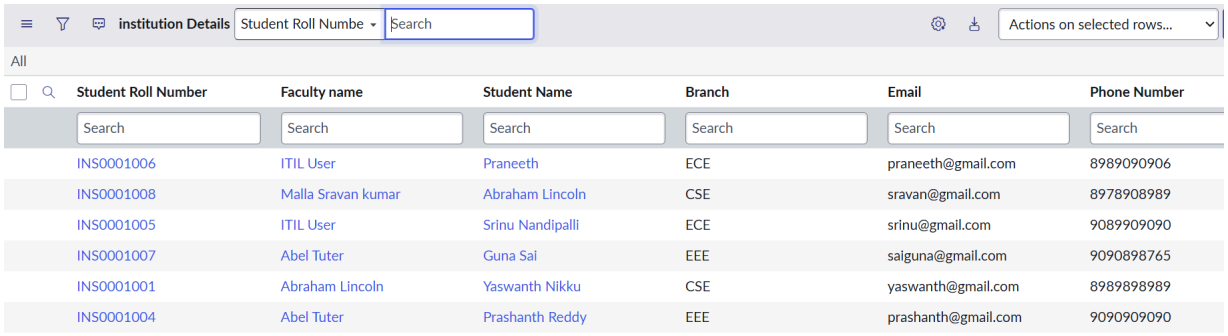




3. Click **Submit**.
4. Add the following fields to the **u_institution_details** table:
    - **Student Roll Number**– Auto Number
    - **Student Name** – Reference- User
    - **Faculty Name** – Reference- User
    - **Branch** – Choice (ECE, EEE, CSE)
    - **Email**– String
    - **Phone Number**– String
    - **Description**– Multi String
5. Save the Table



6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).



Creation of Access Control Lists(Read ACL)

1. In the **Application Navigator**, search for **System Security → Access Control (ACL)**.
2. Elevate “Security_admin” role
3. Click **New** to create a new ACL.
4. Enter the following details:
    - **Type:** record
    - **Operation:** read
    - **Name:** u_institution_details
    - **Active:** true

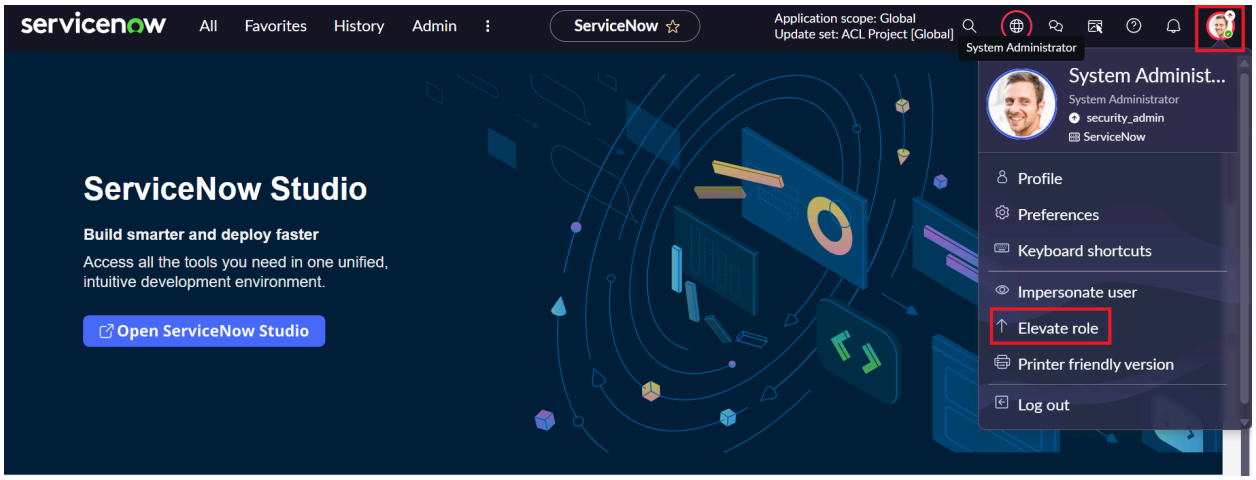
- **Advanced:** true

5. In the **Requires role** related list, add:
   - bb1 (custom role)
6. Add Data condition as **Branch is EEE**
7. In the **Script** field, add the following script:

```
(function () {

  // Allow admin users full access
  if (gs.hasRole('admin')) {
     return true;
  }
  // Allow only EEE branch users to see EEE records
  if (gs.hasRole('bb1')){
     return true;
  }
  // Deny access for all others
  return false;
})();
```

8. Click **Submit** to save the ACL.



Elevate role

Close dialog

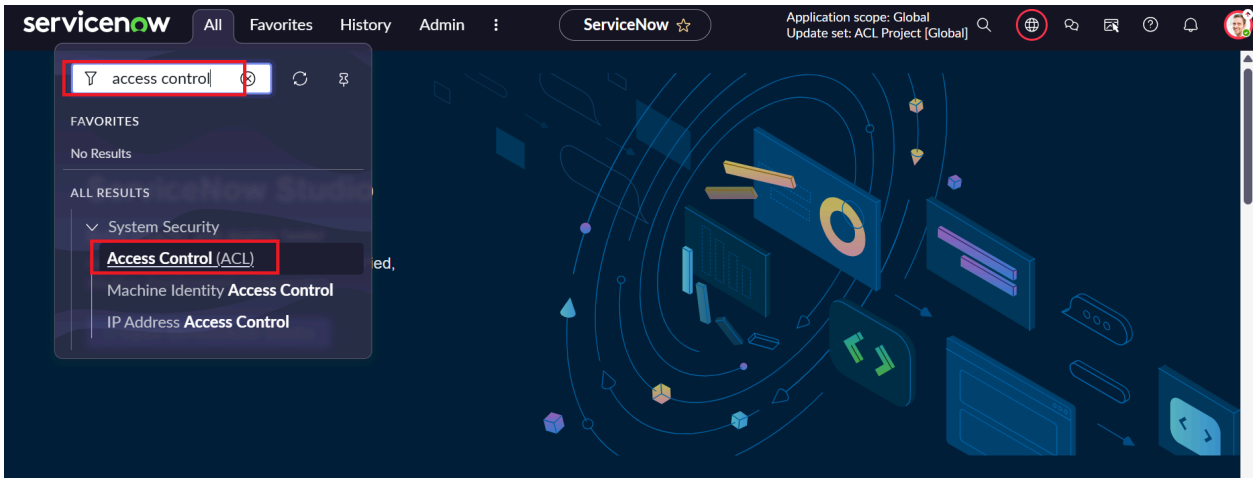
Elevate a role by adding privileges, which end when you log out. Learn more

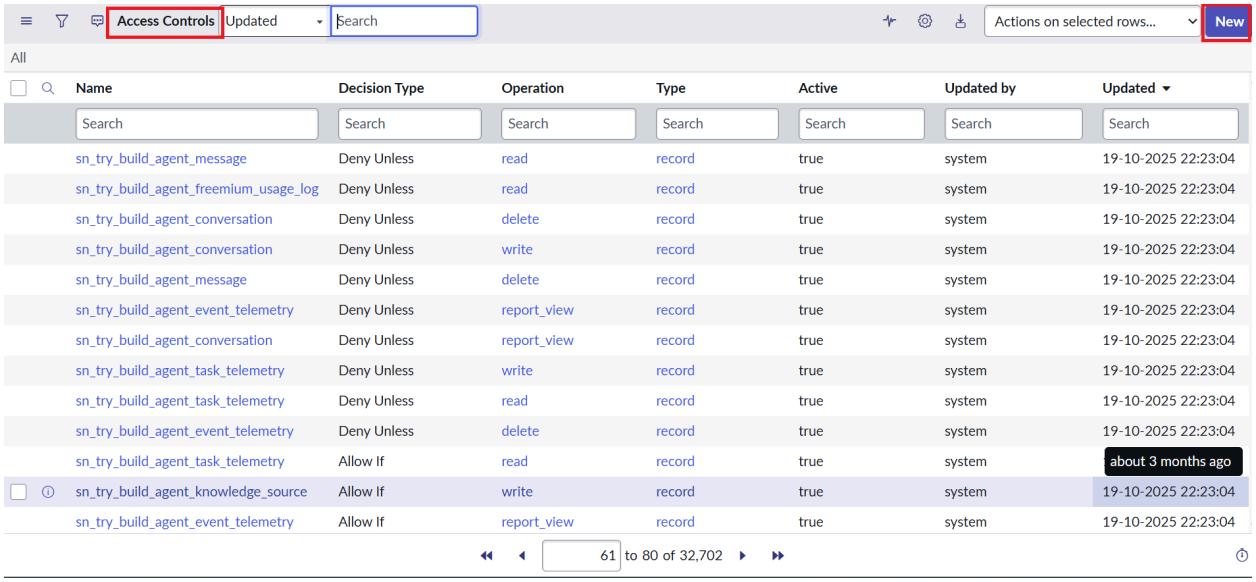
AVAILABLE ROLES

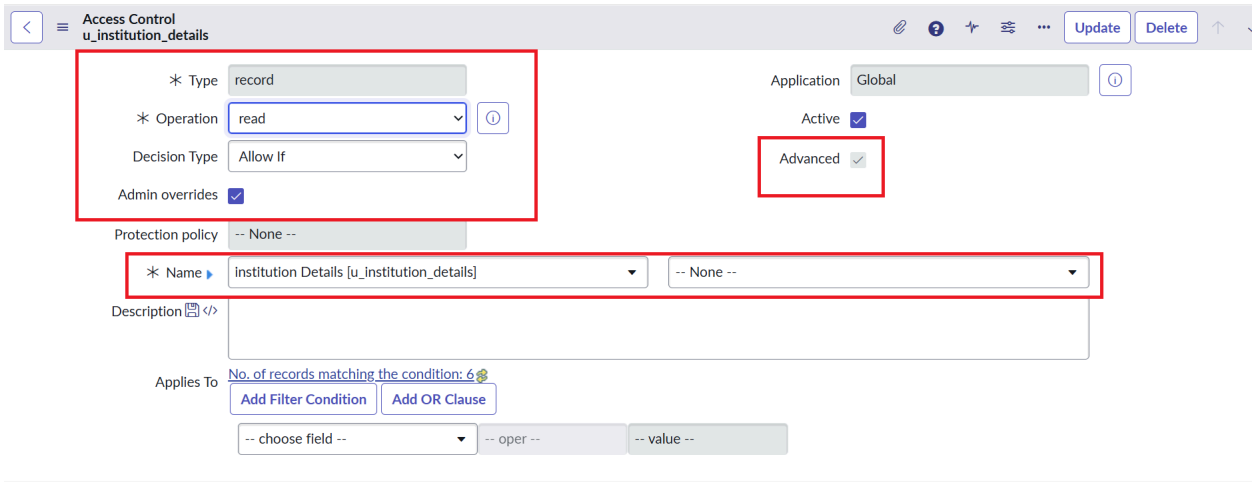
- [x] security_admin

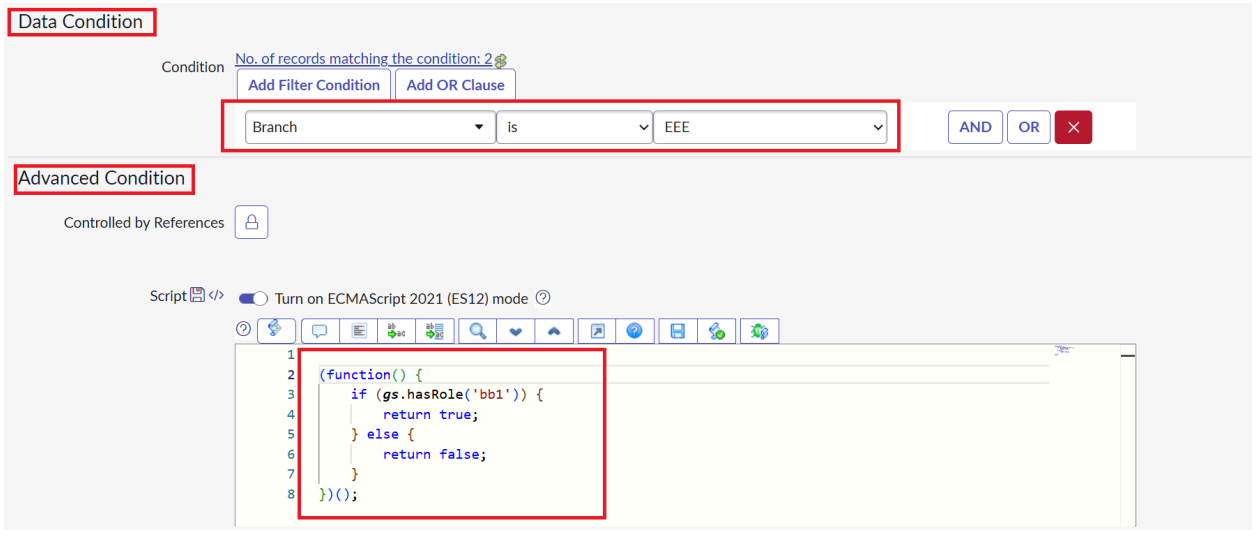
Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel | Update




Access Controls
Updated
Search
Actions on selected rows...
New
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
sn_try_build_agent_message Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_freemium_usage_log Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless write record true system 19-10-2025 22:23:04
sn_try_build_agent_message Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Deny Unless report_view record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless report_view record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Deny Unless write record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Allow If read record true system
about 3 months ago
sn_try_build_agent_knowledge_source Allow If write record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Allow If report_view record true system 19-10-2025 22:23:04
61 to 80 of 32,702

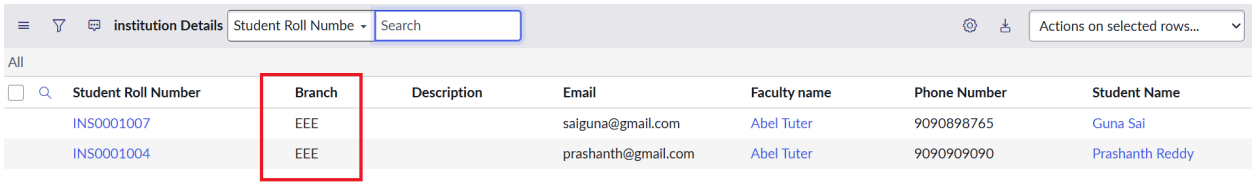

Access Control
u_institution_details
Update
Delete
Type record
Operation read
Decision Type Allow If
Admin overrides
Application Global
Active
Advanced
Protection policy -- None --
Name institution Details [u_institution_details]
-- None --
Description
Applies To
No. of records matching the condition: 6
Add Filter Condition
Add OR Clause
-- choose field --
-- oper --
-- value --


Data Condition
Condition
No. of records matching the condition: 2
Add Filter Condition
Add OR Clause
Branch
is
EEE
AND
OR
Advanced Condition
Controlled by References
Script
Turn on ECMAScript 2021 (ES12) mode
(function() {
if (gs.hasRole('bb1')) {
return true;
} else {
return false;
}
})();

9. Impersonate **EEE users** and open the Student Records list.
10. Copy the table name u_institution_details.list/LIST and paste in the All menu.
11. Impersonate **Admin** and open the Student Records list.

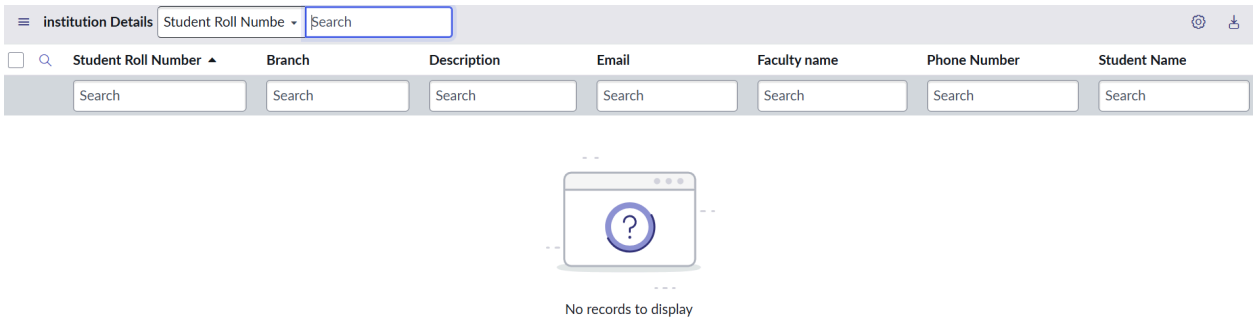
Verify :

- User with **bb1 role** can view **only EEE branch records**

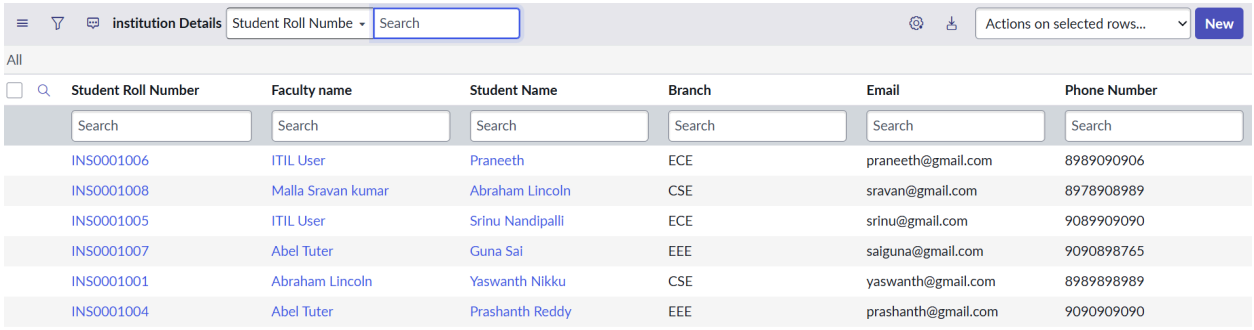


Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE		saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

- User without the role cannot view any records



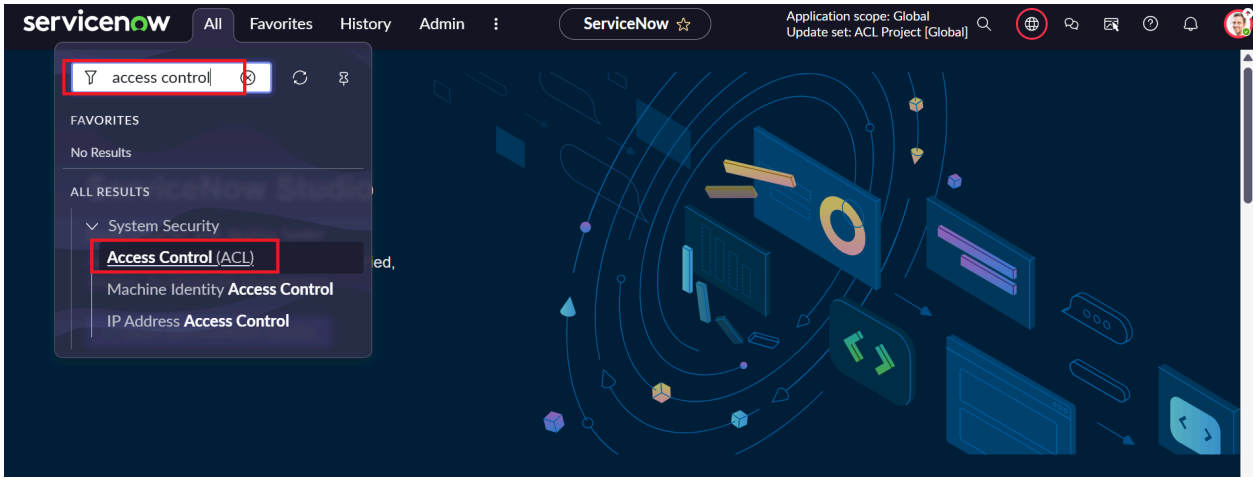
- Admin user can view **all records regardless of branch**

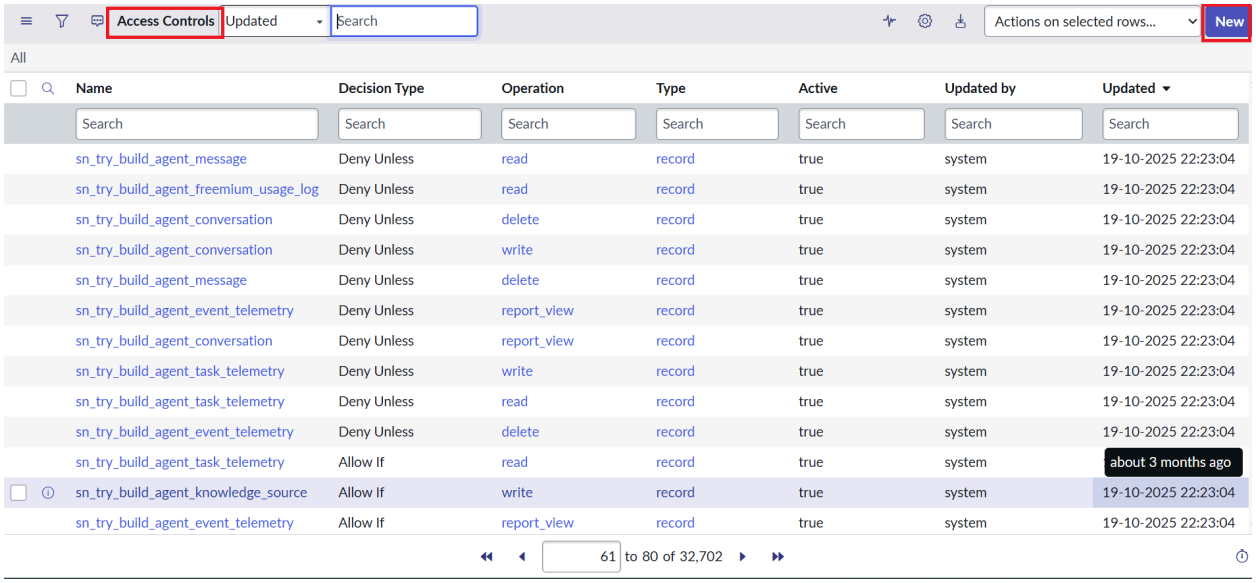


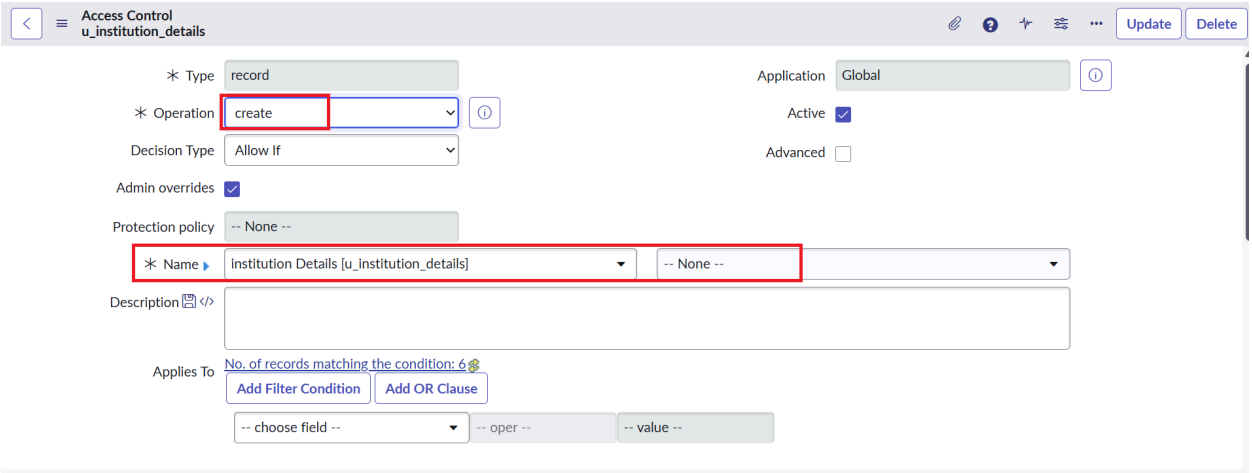
Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number
INS0001006	ITIL User	Praneeth	ECE	praneeth@gmail.com	8989090906
INS0001008	Malla Sravan kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978908989
INS0001005	ITIL User	Srinu Nandipalli	ECE	srinu@gmail.com	9089909090
INS0001007	Abel Tuter	Guna Sai	EEE	saiguna@gmail.com	9090898765
INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989
INS0001004	Abel Tuter	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090

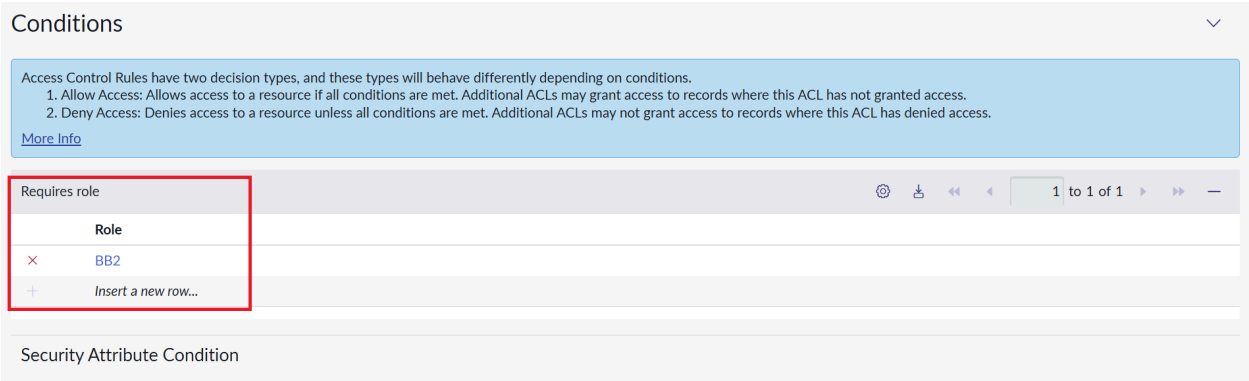
Creation of Access Control Lists(Create ACL)

1. Click **New** to create a new ACL(Create)
2. Enter the following details:
    - **Type:** record
    - **Operation:** create
    - **Name:** u_institution_details
    - **Active:** true
3. In the **Requires role** related list, add:
    - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



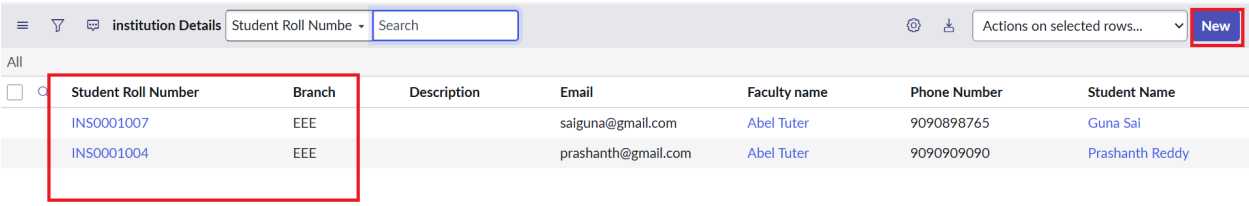






Verify :

- User with **bb1 &bb2 roles** can view **only EEE branch records b New** Button

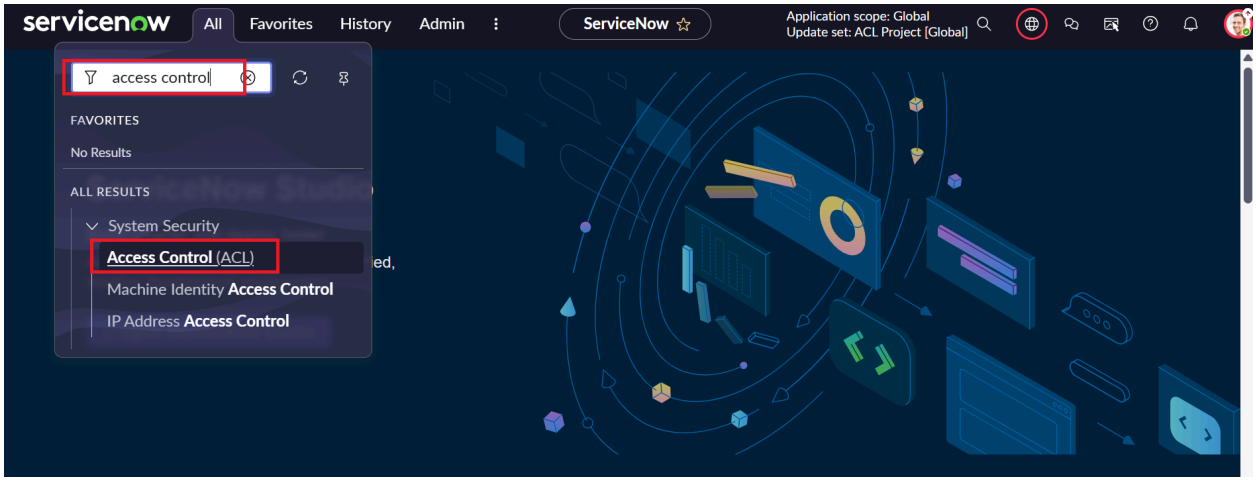


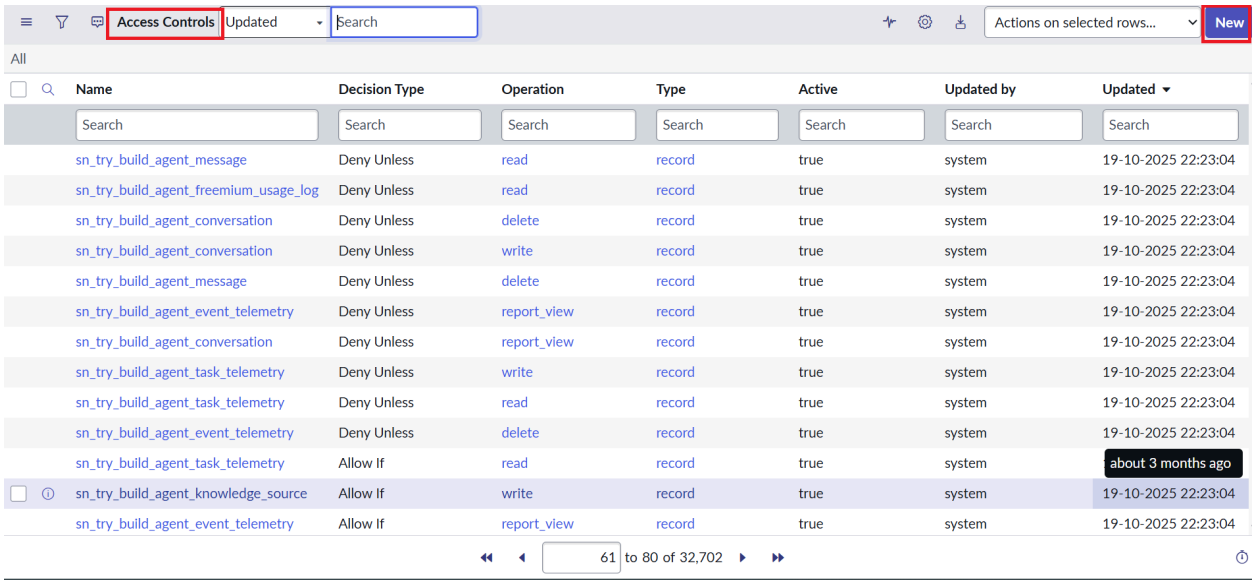
Creation of Access Control Lists(Write ACL)

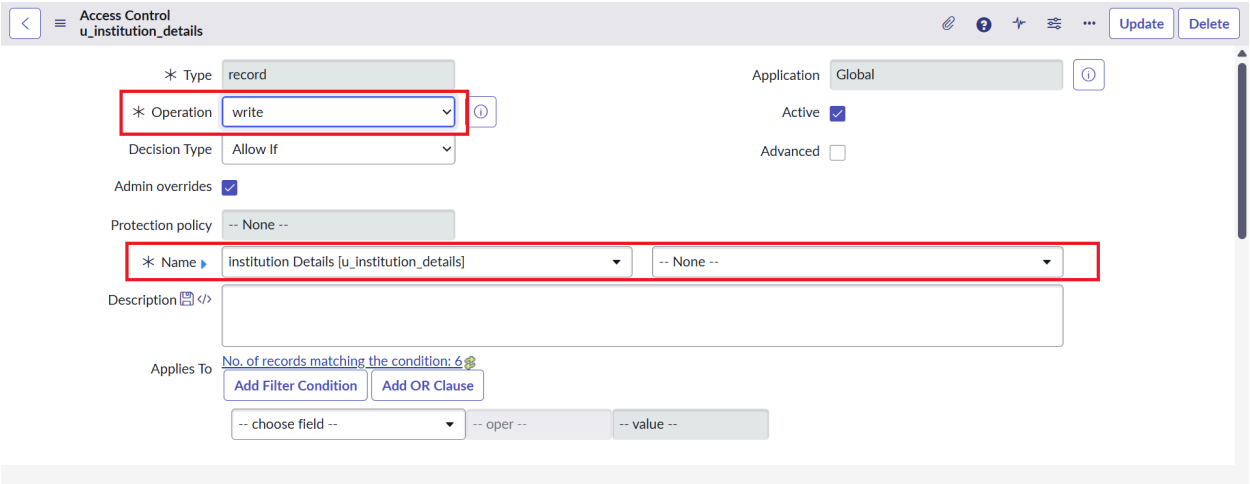
12. Click New to create a new ACL(Write)
13. Enter the following details:
    - Type: record
    - Operation: Write
    - Name: u_institution_details

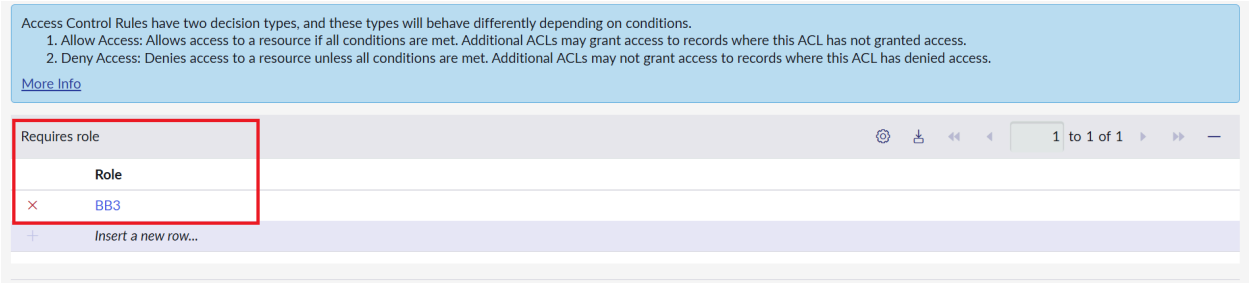
- Active: true

14. In the Requires role related list, add:
    - bb3 (custom role)
15. No need to add any Data condition
16. Save the ACL Record



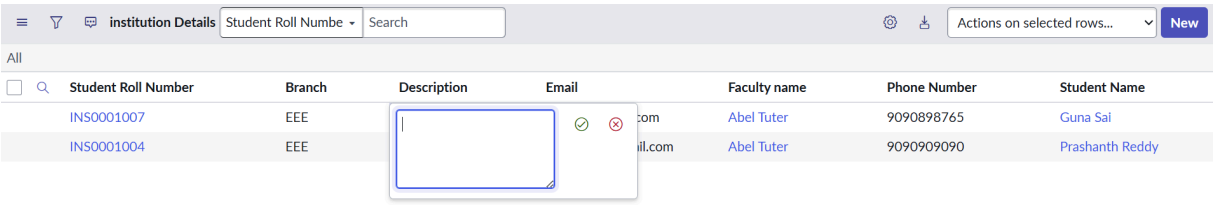


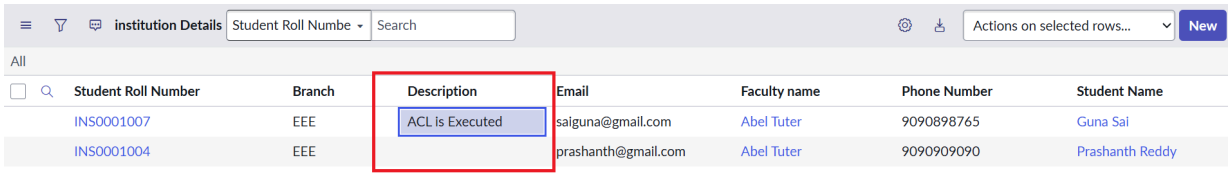




Verify :

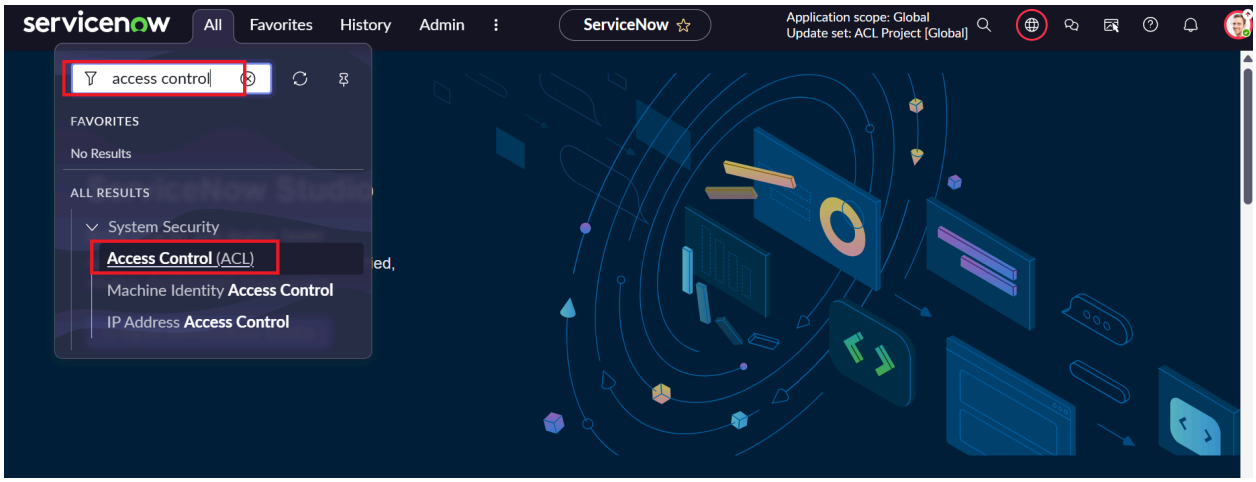
- User with **bb1 &bb2 &bb3 roles** can view and edit the **EEE branch records along with New** Button

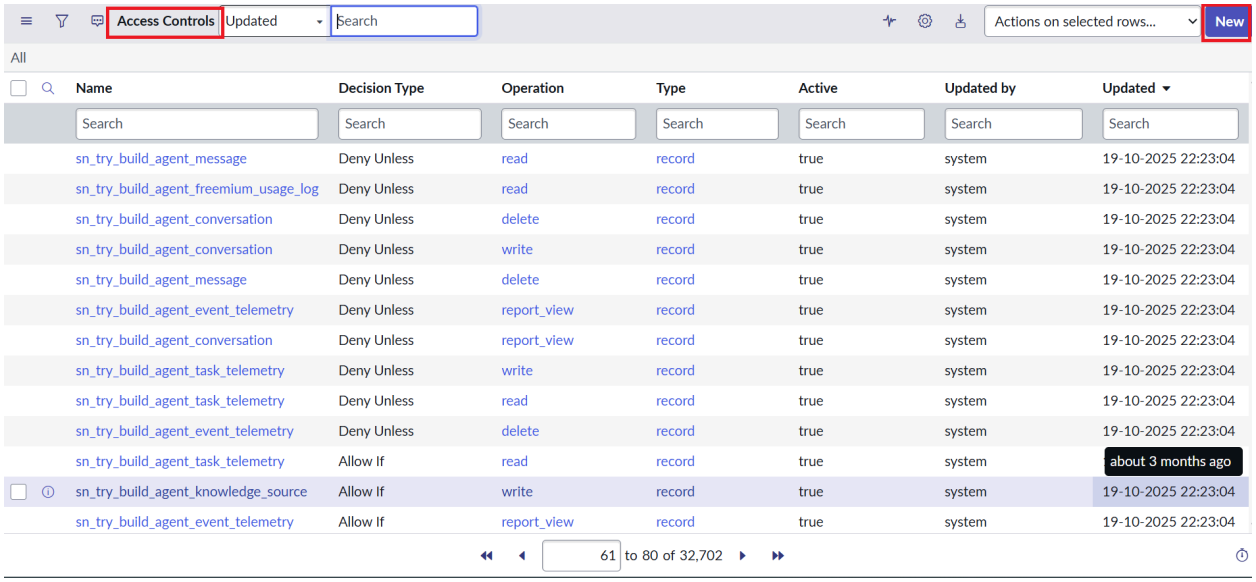


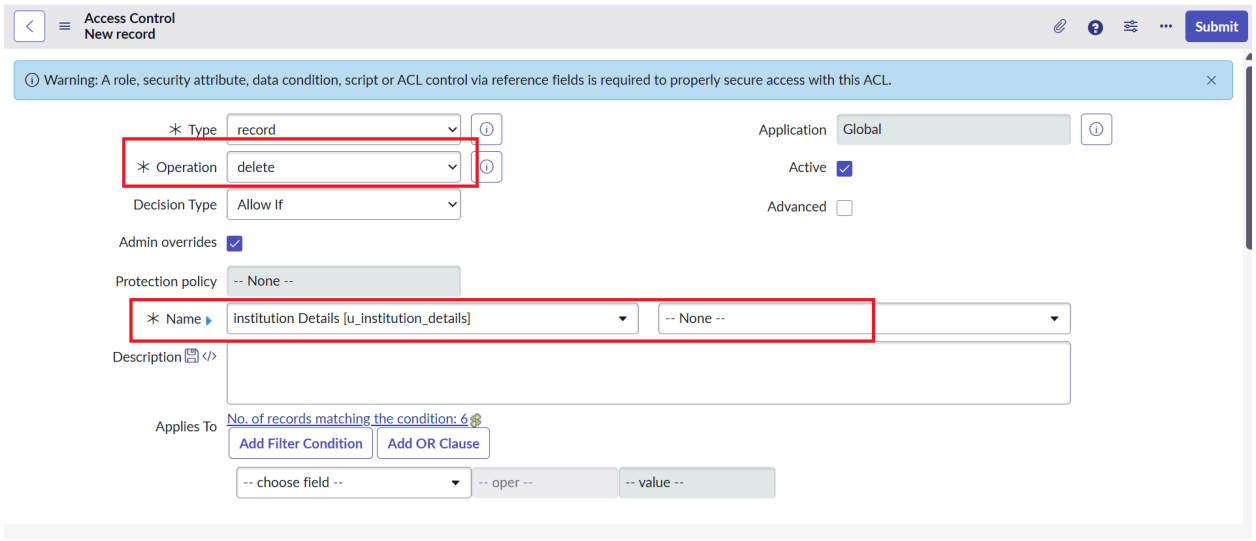


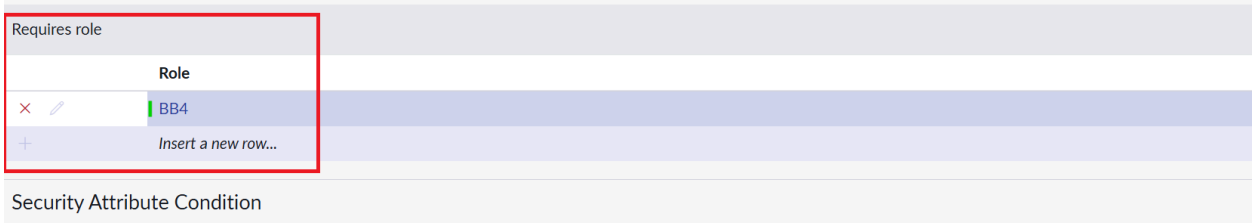
Creation of Access Control Lists(Delete ACL)

17. Click **New** to create a new ACL(Delete)
18. Enter the following details:
    - **Type:** record
    - **Operation:** Delete
    - **Name:** u_institution_details
    - **Active:** true
19. In the **Requires role** related list, add:
    - bb4 (custom role)
20. No need to add any Data condition
21. Save the ACL Record



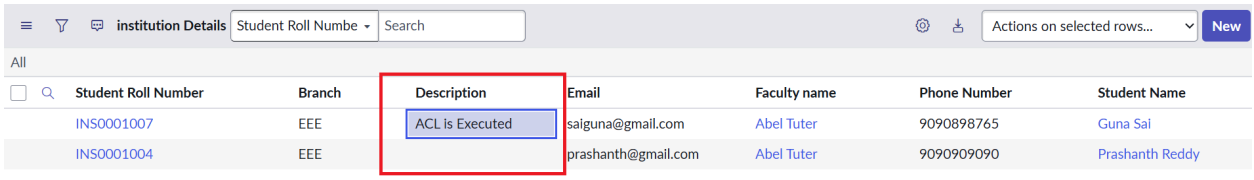


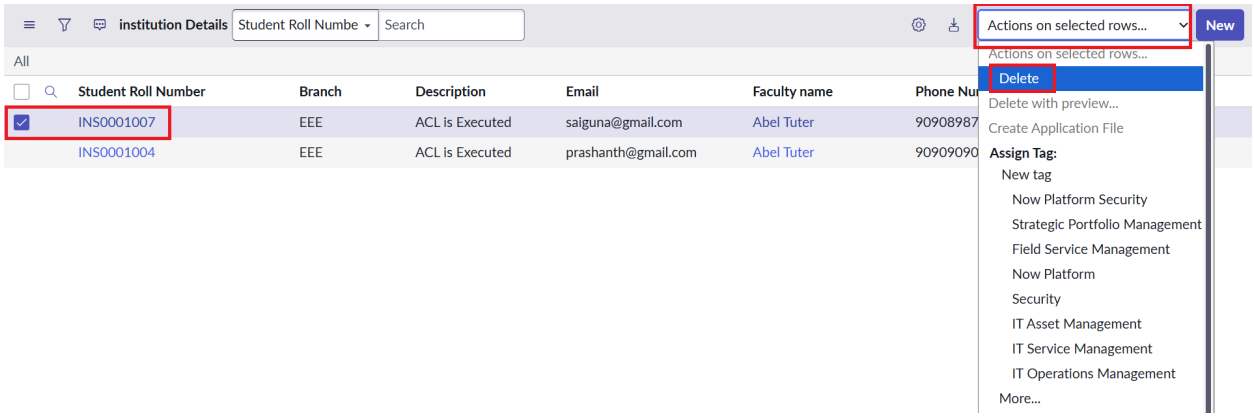


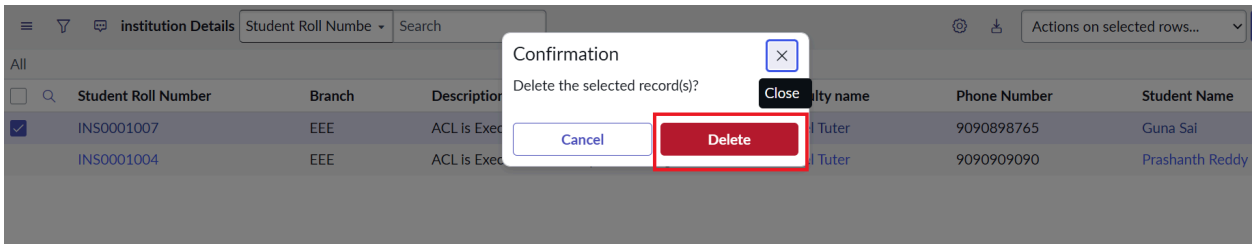


Verify :

- Users with **bb1 &bb2 &bb3&bb4 roles** can see, create and edit along with delete the **EEE branch records.**







Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.